

Data Flow Diagram & User Stories

Date	19 September 2026
Team ID	
Project Name	ShopEZ — Apparel E-Commerce Application
Maximum Marks	4 Marks

1. DFD Level 0: Context Diagram

The Context Diagram shows ShopEZ as a single process, with the two external entities — the Customer and the Store Administrator — and the data that flows between them and the system.

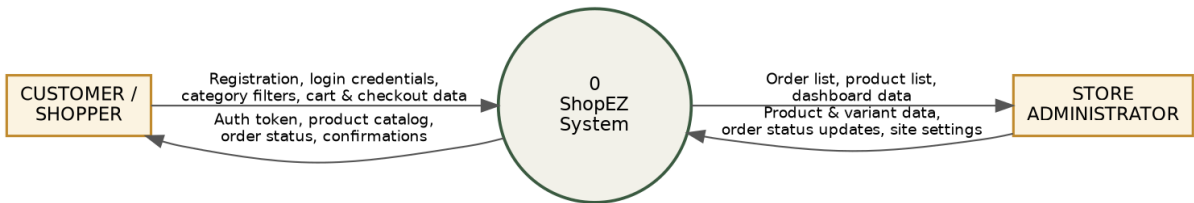


Figure 1: ShopEZ Level 0 context diagram

2. DFD Level 1: Process Breakdown

The Level 1 diagram breaks the single system process into its seven core processes and five data stores, showing exactly how data moves between the customer, the admin, and MongoDB.

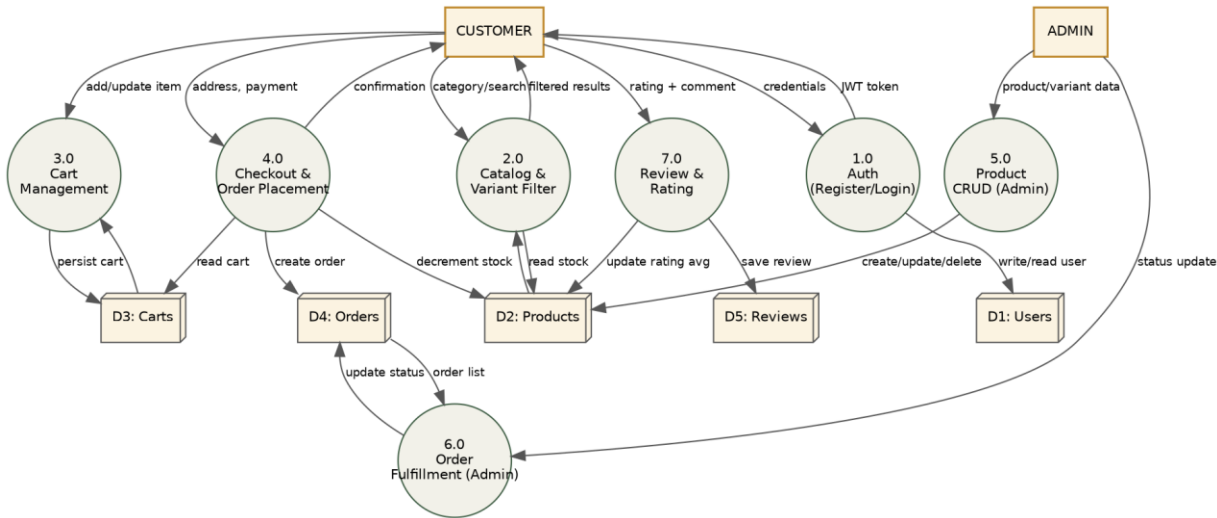


Figure 2: ShopEZ Level 1 data flow diagram

3. Data Flow Diagram Entity and Process Dictionary

Element Identifier	Typology	Description & System Interactivity	Technology Mapping
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Customer	External Entity	Registers, logs in, browses the catalog, manages a cart, checks out, and reviews products.	React Frontend (SPA)
Store Administrator	External Entity	Manages products and variants, updates order status, and configures site settings.	Admin-protected routes
Process 1.0 — Auth	System Process	Handles registration and login; hashes passwords and issues signed JWTs.	Express + bcrypt + JWT
Process 2.0 — Catalog & Variant Filter	System Process	Serves filtered/searched product listings and per-variant stock data.	Express route + Mongoose query
Process 3.0 — Cart Management	System Process	Adds, updates, and removes items in a user's persistent cart.	Express route + Mongoose
Process 4.0 — Checkout & Order Placement	System Process	Re-validates stock, decrements it, creates the order, and clears the cart.	Express controller (transaction-like sequence)
Process 5.0 — Product CRUD (Admin)	System Process	Creates, updates, and deletes products and their variants.	Admin-only Express route
Process 6.0 — Order Fulfillment (Admin)	System Process	Lists all orders and updates their fulfillment status.	Admin-only Express route
Process 7.0 — Review & Rating	System Process	Accepts a review, saves it, and recalculates the product's average rating.	Express controller
D1: Users	Data Store	Stores hashed credentials, role, and profile information.	MongoDB collection (users)
D2: Products	Data Store	Stores product details and the embedded variants array (size, color, stock).	MongoDB collection (products)
D3: Carts	Data Store	Stores each user's current cart items.	MongoDB collection (carts)
D4: Orders	Data Store	Stores placed orders with snapshotted item details and status.	MongoDB collection (orders)
D5: Reviews	Data Store	Stores user ratings and comments per product.	MongoDB collection (reviews)